Migration Control Policy and Parent-Child Separation among Migrant Families: Evidence from China[1]

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The authors declare no conflicts of interest.

[1] Chen acknowledges financial support from the National Natural Science Foundation of China (Grant No. 71773074) and partial financial support from the Key Program of the National Natural Science Foundation of China (Grant No. 72034006). The authors are grateful to Professor Kevin Lang, Jennifer Hunt, Junsen Zhang and Min Lu, as well as to participants at the 2017 Meeting of International Labor Economics Symposium, the 2018 Conference of Migrant Children at Jinan University, the 2019 Summer Institute of CCER at Peking University, and seminar participants at Fudan University for their comments to earlier drafts. All errors are our own.

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Abstract

This paper investigates the effects of a migration control policy in mega cities after 2014 in China on parent-child separation. One of the pivotal initiatives of the policy is to restrict migrant children's access to public education in the city. We exploit variations in the policy implementation pressure across cities and find that the policy contributes to an increased probability of migrant children being left behind. The impact is most prominent among children from families of low socio-economic status and at the compulsory-education age. Little evidence supports that the policy causes a change in the composition of the migrant families in cities.

Keywords: migration control policy, children left behind, rural-urban migrant
JEL classification: I38, J15, K37

1 Introduction

There is a consensus that family environment and investment are key to the accumulation of children's human capital (Bono et al., 2016, Almond et al., 2018, Doyle, 2020). This is especially true when it comes to migrant children (*e.g. Chaudry et al., 2010, Amuedo-Dorantes and Lopez, 2015, Amuedo-Dorantes et al., 2018)*. Therefore, understanding the consequences of stringent immigration policies on the structure of immigrant households is essential. In this study, we aim to examine how a recent migration mandate taking place in 2014 in major Chinese cities has impacted the incidence of parent-child separation among migrant families.

The new migration control policy started in China in 2014, mandated in cities with a central district population of over five million. In recent decades, the rapid rural-to-urban migration flow in China has caused large cities to become overloaded, as the supply of infrastructure and public service can no longer satisfy the increasing population. Due to the pressure of traffic congestion, high housing prices, and environmental pollution in large cities, the central government issued this mandate in 2013, called the "New-Type Urbanization Planning (2014-2020)." It states that cities with a population of more than five million in the central urban districts (*hereafter, mega cities*) should take action to control their population. Since 2014, those cities have started to implement initiatives to reduce the inflow of migrants by shutting down stall hawker bazaars, closing smaller shops in the streets, and moving low-skilled intensive industries out of the city. Among them, the initiative that receives the most criticism is to raise admission standards to public education for migrant children, which places migrant households in a situation to make a trade-off between job opportunities and children's education.

In order to show how migrant parents respond to this trade-off, we first construct a simple conceptual model incorporating an increasing cost to enroll a migrant child in urban schools, expected returns to the child's human capital, and the parents' decision on the family settlement. We postulate that a migrant parent makes decisions on

his child's arrangement and his time allocation sequentially, using an idea similar to backward induction. First, a migrant parent decides whether to leave the city where he works (*hereafter, destination city*) for the sake of children's education, and second, conditional on his decision to stay in the city, he chooses whether to send his child back to rural hometown for education or not. Our model suggests that for low-skilled families that work in cities where the wage is substantially higher than their hometowns, elevated admission cost to public education in cities unexpectedly raises the likelihood of children being left behind in rural hometowns while parents remain in cities.

In the empirical exercise, we collect the goal of migration control policy from local government official documents and create a policy pressure index in the bite of the policy for *mega* cities, those with a population of more than five million in 2013 in the central city district. For each of the *mega cities*, the policy pressure index, defined as the ratio of population in 2013 over the target population in 2020 in the city planning, quantifies how stringent the policy would be implemented in order to achieve the documented goal. The 11 *mega cities* ranked from the highest population control pressure to the lowest are Dongguan, Shanghai, Beijing, Nanjing, Guangzhou, Chongqing, Tianjin, Wuhan, Zhengzhou, Shenyang, and Shenzhen. Our difference-in-differences identification exploits variations in the policy pressure across cities and uses so far the largest migrant data in China, the Chinese Migrant Dynamic Monitoring Survey (CMDMS) from 2011 to 2016 conducted by the China National Health and Family Planning Commission.

Our primary empirical task is to examine the policy effect on parent-child separation, specifically, the scenario when migrant parents leave their children behind at their hometowns due to the limited access to public education in the destination cities. Focusing on a sample of migrant children at the age of primary school or middle school (the compulsory education stage), we find that the 2014 population initiative causes a remarkable share of migrant children being left behind: every one percentage point increase in the policy pressure leads to 0.43 percentage points increase in the probability of children being left behind. To put this estimate in perspective, given that the

proportion of left-behind children is 0.384 prior to the policy among the *mega* cities, our result indicates that one percentage point increase in the policy pressure raises the proportion of children left behind by 1.12 percent. The path of coefficients from a more parsimonious event study exhibits a persistent and sizable policy effect immediately after the policy enactment, but no indications of differential trends in the probability of children being left behind across cities prior to the policy.

We perform several mechanism analyses to ascertain that our estimates primarily reflect the impacts from the education-based migration control policy and are not driven by other concurrent changes in the macro-economic environment. Two natures of the policy guide our analyses. *First*, the migration policy is skill-biased, as it aims to control the migration inflow and upgrade the skill level of the migrant labor force by phasing out low-skilled migrants. Thereby, families of low social-economic status are expected to be hit disproportionately. We find that the policy impact is the most pronounced for children from less-educated or low-income families, who are more than three times as likely to be left behind as their counterparts from families of higher social-economic status.

Second, the elevated admission to public schools should mainly affect children at compulsory-education stage. To evaluate differential policy effects across education stages, we explore the heterogeneity in exposure to the policy across children's ages. The policy effect shows up primarily for children in the compulsory education period and less prominently for children in the pre-school stage or beyond middle school. To dive deeper, we further construct an index measuring the public school enrollment barrier for migrant children based upon the detailed requirements from the official documents in each city on admission to public primary or middle schools. The results confirm with the above hypothesis that the elevated admission standard only matters for children at the compulsory-education stage. Lastly, we rule out a series of alternative explanations, such as air pollution in large cities, rapidly increasing housing prices, high-speed railway developments, and improved educational resources in rural areas.

Given that the CMDMS data is repeated cross-sectional, another threat to the va-

lidity of our difference-in-differences design is compositional changes in migrant families in cities over time. If migrant parents who seriously value their children's education leave cities with a more stringent policy for cities with a more generous one or rural hometowns, our baseline results about parent-child separation may carry this composition effect.

To test this compositional change, we focus on a sample of migrant families. The empirical findings do not endorse remarkable compositional differences in education or family structure. For example, we do not find a significant reduction in the share of low-skilled migrants or migrants with at least one child at the compulsory school stage. We further investigate the policy impact on the labor allocation with spouses. Specifically, the increase in school enrollment requirements may induce one parent to return home to take care of their child, a scenario we refer to as the spousal separation. The point estimates on spousal separation are positive but not statistically distinguishable from zero, indicating that minimal policy impact. These evidence implies that the majority of low-skilled migrants forgo their companions with children for better job opportunities and earnings in *mega cities*, as the *mega cities* pay extra wage premium (Xing and Zhang, 2017).

Our study contributes to the existing literature in several aspects. *First*, it stands with a large literature that evaluates immigration policy. The migration decision "is mainly driven by structural factors, such as labor market imbalances, inequalities in wealth, and factors on which migration policies have little or no influence" (Czaika and De Haas, 2013). Policymakers pursuing a short-term political goal may enact mandatory initiatives in a way that overlooks immigrants' strategic responses to these policies, leading to undesirable consequences. These consequences include, but are not limited to, increasing unemployment (Amuedo-Dorantes et al., 2013, Wang, 2019, Amuedo-Dorantes et al., 2021), more crime activities (Gathmann, 2008, Miles and Cox, 2014, Mastrobuoni and Pinotti, 2015, Freedman et al., 2018), and less insurance and health care utilization (Watson, 2014, Page and Polk, 2017), and reduction in school attendance of immigrant children with unauthorized migrant parents (Capps et al.,

2007, Chaudry et al., 2010, Amuedo-Dorantes et al., 2013, Amuedo-Dorantes and Lopez, 2015, 2017, Amuedo-Dorantes et al., 2018, Dee and Murphy, 2020).

In our study context, for the goal of restricting migration inflow, especially the low-skilled labor inflow, the local governments in *mega cities* in China implemented a policy attempting to affect adults' migration decisions by limiting their children's access to public education rather than through regulating structural factors in the labor market. Whether this policy is successful and the unanticipated costs associated with it remain still unclear yet.

Second, to the best of our knowledge, this paper is one among the few that attempt to explore how internal migration policy in developing countries affects parent-child separations in migrant families. Our study echoes recent studies assessing the impact of immigration enforcement on immigrant marriage patterns (Amuedo-Dorantes et al., 2020, Bansak and Pearlman, 2022) and living arrangement among immigrant families (Amuedo-Dorantes et al., 2013, Amuedo-Dorantes and Lopez, 2015, Amuedo-Dorantes et al., 2015, Dreby, 2015, Amuedo-Dorantes and Arenas-Arroyo, 2019) in developed countries. Compared to international immigration, migration from the rural to urban areas within a country is virtually free. It thus is in even larger volumes as a consequence of the substantial rural-urban economic gap. Concerns linger to what extend the lesson learned from international immigration policies could guide the migration policies aiming to control the (internal) inflow of rural migrants to urban within developing countries.[1]

China, as the largest developing country in the world, creates an ideal context. China's economic reform and rapid urbanization over the past three decades have led to a massive, unprecedented rural-to-urban migration, resulting in a substantial population of children separated from their migrant parents (Chan and Ren, 2018). The 2018 joint report by the Women and Children Committee in State Council, the UNICEF

[1]Unlike harsh actions such as deporting undocumented cross-border immigrants, internal migration initiatives usually disincentivize the inflow of rural immigrants by limiting their access to public welfare and urban services such as access to health insurance and education. The welfare loss of migrants families associated with internal migration policies in developing countries is often overlooked.

and the China National Bureau of Statistics estimates that the population of children under 18 years old separated from at least one migrant parent has reached as high as 87.60 million, accounting for almost one-third of children in China.[2] The lessons from China could also help guide population policies in other developing countries, which may encounter similar migration problems during their rapid urbanization.

Last, the parent-child resulting from the immigration policy has attracted increasing concerns owing to the detrimental effects of parental absence on children's academic performance and mental health (Meyerhoefer and Chen, 2011, Zhang et al., 2014, Zhao et al., 2014, Meng and Yamauchi, 2017, Antman, 2013), and also on their future labor market outcomes in the long run (Lyu and Chen, 2019). The substantial population of children living in the absence of their parents in China may impede human capital accumulation and economic growth sustainability. The central government of China announced a directive in 2016, emphasizing the care and protection of left-behind children in rural areas, introducing actions to reduce the population of left-behind children. Our study renders an important policy implication that providing migrant children in cities with, rather than isolating them from, access to public education resources is an effective way to solve the issue of left-behind children.

This paper is organized as follows. Section 2 provides the institutional background on the migration control enforcement policy in *mega cities* in China. Section 3 develops a theoretical model. Section 4 describes the data and the estimation methodology. Sections 5 and 6 report the baseline empirical results on parent-child separation and compositional change of migrant families, respectively. Section 7 concludes the paper.

[2]Among all children separated from their migrant parents, there are 40.51 million left-behind children in their rural hometowns, 28.26 million left-behind children in their urban hometowns and 18.84 million left-behind children who migrate to other cities where they do not have *hukou* residence.

2 Background - Migration Control Enforcement in *Mega Cities* in China

In this section, we provide the institutional background of migration control enforcement in *mega cities* in China. We first start with the general description of rural-to-urban migration and discuss the incentives of migration control enforcement in *mega cities*. Then, we focus on the education-based migration control policy starting in 2014 and elaborate on the elevated school enrollment restrictions in *mega cities.*

2.1 Migration Trend Towards *Mega Cities*

Over the past three decades, China has witnessed the most significant internal migration in human history. During the first decade of the 21st century, the migrant population doubled and reached 221 million in 2010. The growth rate of migrant population slowed after 2010 but maintained a significant annual growth rate as high as 3.44% from 2010 to 2014. Until 2014, the internal migrant population reached 253 million, accounting for 18.5% of the total national population (NBS data).[3] Moreover, the pattern of labor migration has transited from rural-to-urban migration in the 1990s to small-city-to-large-city migration in the 21st century, as larger cities offer better job opportunities and pay higher wages to migrants (Xing and Zhang, 2017, Liu, 2014). In *mega cities*, such as Beijing, Shanghai, Guangzhou, and Shenzhen, more than 40 percent of their populations are migrants, whereas the proportion of migrants in the national population is 18 percent according to the 2015 census data (Yang, 2016).

[3]The NBS defines migrants as those who move to live for more than six months in cities where they do not have local *hukou*. Hence, the permanent migrants who manage to obtain *hukou* in the destination cities are counted as local residents. In the 2020 census data, the migrant population reached 376 million, though how much of the statistics is due to the increase of the actual migrants or the measurement issue in previous survey is still under debate.

2.2 Migration Population Control Policy in *Mega Cities*

The rapid migrant inflow has caused *mega cities* to become overcrowded, as the supply of infrastructure and public services can no longer accommodate the increasing population. The so-called larger-city problems, such as high housing prices, traffic congestion, and environmental pollution, have attracted increasing concerns. In 2013, the National Development and Reform Commission issued the "National New-Type Urbanization Planning (2014-2020)", which urged *mega cities*, defined as those with a population of over five million in the city central district area, to "rigidly control the population".[4] Each *mega city* began to set its population target by 2020, and the success in population control has become one of the most crucial performance evaluation indices of local governments. Moreover, the Chinese economy started to transit from low-skilled-intensive production to high-skilled-intensive production. *Mega cities* also aim to seize this opportunity to optimize the skill composition of their labor force by embracing more high-skilled workers and restricting influx of low-skilled migrants.

Local governments implemented various initiatives to achieve the population goal, including shutting down stall hawker bazaars, closing smaller shops in the streets, moving some low-skilled intensive industries out of the city, and imposing stringent school enrollment requirements for migrant children. Meanwhile, it became more difficult for migrants to obtain identities related to the local public services, such as *hukou*, a legal residence permit (*Ju1 Zhu4 Zheng4*) or identity-related points accumulation in the point-based system.[5] The intensity of these policy implementations increases with population control pressure.

[4] http://ghs.ndrc.gov.cn/zttp/xxczhjs/ghzc/201605/t20160505_800839.html.

[5] Starting from 2014, more and more cities started to implement a points-based system, whereby points can be accumulated according to a migrant's education level, skill specialization, and occupation, as well as social security and tax payment. A migrant's right to access public service in the city is based on their points. However, it is almost impossible for low-skilled migrants to obtain the necessary points.

2.3 Education Related Migration Control Policies in *Mega Cities*

In this subsection, we will separately discuss the education enrollment policy for migrant children at the compulsory education stage, including elementary and secondary schools, and beyond the compulsory education stages, such as pre-schools, high schools, or college and above.[6]

2.3.1 School Enrollment Requirements for Public Schools at the Compulsory Education Stage in *Mega Cities*

The educational rights of migrant children in the city are closely related to China's *hukou* system. Before 2001, migrant children without local *hukou* were not entitled to free compulsory education in the city, as the budget for public schools was primarily from local government and not transferable across cities. Local governments lacked the incentives to provide education to migrant children. Public schools at that time charged "out-of-district" fees and required a dossier of documents and certificates which many migrant parents were unable to provide. As a result, privately operated migrant schools in the cities started as a market-based response to the educational needs of migrant children. However, the poor-quality education offered by migrant schools becomes a concern since school owners with a limited budget have to minimize operating expenses, and are unable to hire experienced teachers and purchase necessary teaching equipment. These schools are subject to constant closures and changes of location due to the lack of a clear legal status within China's educational system. In addition, migrant schools and their students are "not in the system" and face substantial institutional barriers (Chen and Feng, 2013).

In 2001, the central government issued a directive to encourage destination city governments to take responsibility for the compulsory education of migrant children

[6]For details about the educational rights of migrant children in China, see a literature review (Chen et al., 2019).

and to enroll them in public schools. Gradually, destination cities have started to allow students to enroll in public schools if their parents meet certain admission requirements and there are seats available. The proportion of migrant students in public schools in urban cities has subsequently increased since then. According to a State Council Education Committee report in 2014, around 12.50 million migrant children were enrolled in urban schools, with 79.5% of them studying in public schools (Xu and Zhang, 2016). However, large cities have an inadequate supply of public schools to satisfy the educational needs of all migrant children. Migrant children who are at compulsory education age (usually 6 to 15 years old) and can not enter public schools have to enroll in migrant schools if their parents want to keep them in the city. By 2014, around 2.6 million migrant children were studying in migrant schools (Yang, 2016). Hence, public schools and private migrant schools are the two main school types to provide elementary and middle school education for migrant students.[7] We now discuss the policies regarding these two types of schools, respectively.

The enrollment requirement in public schools has elevated, to varying degrees, in all *mega cities* since 2014, depending on the local government's determination and policy pressure to control the population. For cities with more intensive population control pressure, such as Beijing and Shanghai, the policy implementation was expected to be more stringent after 2014. In Shanghai, before 2014, the school enrollment only required that at least one parent should have a temporary residence permit (*zan4 zhu4 zheng4*), which could be obtained if a parent has been employed in Shanghai for more than six months. Since 2014, the school enrollment requirement has increased significantly and now requires a legal residence permit (*ju1 zhu4 zheng4*), which requires that either parent must pay social security for at least six months. Thus, migrant parents either have to be formally employed in firms that pay above 1000 RMB per month for their

[7]In many cities, such as Shanghai, migrant schools are only available for primary education, but not for middle school education.

social security or need to pay this money out of their own pockets.[8] In Beijing, it requires migrant parents to submit more than 20 certifications after 2014 rather than five before 2014 to satisfy the public school enrollment standard. Those certifications include, but are not limited to, proof of their living and employment status, minimum employment duration within the district, proof of tax and social security payments, and proof of no custody in their hometown.

Cities with less population control pressure, such as Nanjing and Wuhan, also started to require legal residence permits, social security payments, or legal business licenses. Even though some officially required documents are the same, cities with different population pressure implement these policies with different efforts. For example, the actual policy implementation in Nanjing and Wuhan is not as strict as in Beijing and Shanghai. Cities with slack population growth margins can artificially relax the procedures in obtaining "*ju1 zhu4 zheng4*". Meanwhile, schools in such cities could have the flexibility to enroll students who do not meet the requirement if the schools have seats available after enrolling the entitled students.

2.3.2 School Enrollment Requirements for Migrant Schools at the Compulsory Education Stage in *Mega Cities*

There are also substantial variations in the migrant school system across *mega cities*. It has also become dramatically challenging for migrant children to study in migrant schools in cities with intense population control pressure, such as Shanghai and Beijing. For example, Shanghai implemented the most migrant-friendly policy before 2014 among *mega cities*, in terms of accommodating migrant children's educational needs. The government fully financed migrant schools, and students in all migrant

[8]The only exception is to have a flexible employment certificate, which applies to some specific and local urgently needed occupations, such as carers in hospitals or nursing homes, and some employees in the agricultural sector. However, this flexible employment certificate is difficult to obtain and requires a large number of documents. There are also many hidden rules, and the difficulty of obtaining these certificates increases with the population control pressures in local districts.

schools were also entitled to free education.[9] However, since 2014, the new school enrollment policy has been applied not only to public schools but also to migrant schools. Schools not subjected to these requirement changes are elite private schools or international schools, which charge tuition fees that are far beyond the affordability of most migrant families. Thereby, migrant students who cannot satisfy the new school enrollment requirements have no other options to study in Shanghai. According to a survey conducted by the School of Social Studies at Shanghai University, with the sudden increase in school enrollment requirements, the number of migrant children enrolled in compulsory-education-stage schools decreased by 38,000 in Shanghai in the fall of 2014. In Beijing, the case is slightly different. Migrant schools charge tuition fees, and there are no explicit restrictions on enrollment into the migrant schools. However, migrant schools have a very high risk of being shut down due to various policy reasons, such as land expropriation or housing demolition. Since 2014, local governments have had strong incentives to close more migrant schools with the intense migration control pressure.

Migrant schools in cities with less population control pressure, such as Nanjing and Wuhan, do not have enrollment requirements and charge tuition for around 2,000 to 6,000 RMB per year. For students who qualify for public schools but remain in migrant schools, the government gives a subsidy of around 1,000 RMB per year for each qualified student. As more students enter public schools, the number of migrant schools in these cities gradually decreased. The exception are Guangzhou and Shenzhen, where around half of the migrant students enroll in migrant schools. Migrant schools generally do not have enrollment requirements and charge tuition from around 5,000 to 20,000 per year. Although after 2014, Guangzhou and Shenzhen started to control the number of students in migrant schools, the large capacity of migrant schools can still accommodate a large population of migrant children. Therefore, the admission standard to migrant schools was not elevated as much as in Beijing and Shanghai.

[9]For details on the development of migrant schools in Shanghai before 2014, please refer to Chen and Feng (2017).

2.3.3 School Enrollment Requirements Beyond Compulsory Education in *Mega* Cities

Beyond the compulsory education stages, the destination cities do not provide many public education opportunities for migrant children. In *mega cities*, the resource of public kindergartens is far less than that for elementary schools. Most migrant families have to pay for private kindergartens, and their children mostly enter unauthorized and low-quality migrant kindergartens or care centers.

For education beyond middle schools, the policy has not changed much after 2014. Even before 2014, it was almost impossible for general migrant students to take the high school or college entrance exam in the cities where they did not have local *hukou*. Most migrant students had to return to their hometowns for high school education. Even though some provinces have started to open access to high school and college entrance exams to migrant families in recent years, most of the access is restricted to families who earn enough points in the point-based system. Wu and Zhu (2014) document the requirements of the high school or college entrance exam for migrant students in various cities and conclude that the reform only covers a tiny proportion of migrant students.

3 Conceptual Framework

3.1 Setup

We present a simple human capital investment model to demonstrate the potential impact of school enrollment requirement changes in *mega cities* on a migrant family living arrangement. The model incorporates the cost to enroll a child in urban schools, parental expected returns to the child's human capital accumulation, and the parent-child living arrangement. To make our model illuminative but straightforward, we focus on migrant families in the city and assume that each family has a single parent and

only one child.[10]

Similar to Shah and Steinberg (2017), a migrant parent maximizes the total family utility, which consists of the utility flows from consumption in the first-period and the returns to his offspring's human capital in the second period. We can phrase the migrant parent's utility maximization problem as

$$\underset{t\in[0,T]}{\text{Max}} \quad lnC + \beta_0 lnQ, \tag{1}$$

$$\text{s.t} \quad c + e\theta(X, d_c) = E(w_c|X)(T-t) + E(w_r|X)t, \tag{2}$$

$$Q = \kappa_0 S^{\kappa_1} A^{\kappa_2}[e(T-t) + (1-e)t]^{\kappa_3}, \tag{3}$$

where β_0 is the migrant parent's utility discount coefficient. Equation (2) is the family's budget constraint in the first stage, which assumes a parent deposits no savings in the first period and retires in the second period. We assume that a migrant parent earns a wage of $E(w_c|X)$ if working in the city and a wage of $E(w_r|X)$ if working in his rural hometown. Both wages are a function of X, a set of characteristics such as skill level and education attainment. The wage in the city is higher than in the rural areas, i.e. $E(w_c|X) > E(w_r|X)$. A parent can allocate his entire time endowment T to work either in the city or in the rural hometown or allocate a part of T to work in the city and the rest to work in the rural areas. For simplicity, we assume a zero commuting cost between rural and urban areas.

We assume that the quality of education in the city is superior to that in the rural areas.[11] To capture the urban-rural education quality gap, we model the rural

[10]Though the model is based on migrant parents' decision, it can also be extended to explain the potential migrants who currently live in the rural home and are planning to migrate. In doing so, the model also has to consider different migration destination alternatives.

[11]Chen and Feng (2017) find that around 80 percent of migrant parents think that the quality of public schools in Shanghai is better than the quality of schools in their hometown, and around 50 percent think that migrant schools in Shanghai are better compared to those in their hometown. Lai et al. (2014) compares the math scores of students in migrant schools in Beijing with those in rural public schools in Shaanxi and finds that migrant schools in Beijing outperform rural public schools in Shaanxi.

education quality q_r as a proportion of the urban education quality q_c: $q_r = \gamma q_c$, where $0 < \gamma < 1$.[12] If a child is sent back to a school in the rural hometown, the education cost is normalized to 0 because of the free compulsory education law in China. To enroll the child in a public school in the city, the parent needs to pay an effort $\theta(X, d_c)$, including, but not limited to, the cost of obtaining the required legal documents, social security payment, and out-of-district extra tuition fees.

The migrant parent's cost $\theta(X, d_c)$ depends on his characteristics X and the policy pressure in controlling the city population growth, d_c. When the local city government faces intensive pressure and thus implements a more stringent school enrollment standard for migrant children, $\theta(X, d_c)$ increases accordingly, i.e., $\frac{\partial\theta(X,d_c)}{\partial d_c} > 0$. To capture the fact that the migrant control policy favors more high-skilled workers and hits children from low-skilled families the most, we assume that $\frac{\partial\theta(X,d_c)}{\partial X} < 0$, implying that the barrier to the public education decreases as the parent's skill level increases. The parent spends his earnings on family consumption c and the child's education cost $e\theta(X, d_c)$, where e is a binary variable indicating the child's living arrangement that equals to 1 if the child studies in the city and 0 otherwise, and $\theta(X, d_c)$ is the cost to enroll the child in a public school in the city.

Equation (3) is a variant of the well-known Ben-Porath human capital investment function (Ben-Porath, 1967), where a child's human capital production function Q consists of three inputs: the year of schooling S which depends on the school quality: $S = S(q)$ where $q \in \{q_c, q_r\}$; the child's innate ability A; and parental time involvement $[e(T - t) + (1 - e)t]$. Parameter κ_0 is the Hicks neutral-efficiency parameter for human capital production. Parameters $\kappa_1, \kappa_2, \kappa_3 \in [0, 1]$ measures the efficiency of each input. Without loss of generality, we assume that if the parent sends his child back to the rural hometown and spends time t working in that area and $T - t$ in the city, then the parental time involvement is t.

[12] In practice, it is possible that the parent may send his child to a smaller city for education, rather than back to rural hometown. Our model can incorporate this case by assuming that education costs are lower in cities with no migration control policy.

3.2 Parent's Optimal Strategic Behavior

We postulate that the migrant parent makes decisions on his child's arrangements and his time allocation sequentially, using an idea similar to backward induction. First, conditioning that the parent has chosen his child's education arrangement (e), he makes his decision on the time spent with the child, t.

When $e = 1$: The migrant parent keeps his child with himself in the city and enrolls his child in the city's school education system. In this scenario, it is apparent that his best time allocation strategy is to work in the city. Therefore, $t^*_{e=1} = 0$. The migrant parent's utility would be

$$U_{e=1} = ln(E(w_c|X)T - \theta(X, d_c)) + \beta_0 ln\kappa_0 + \beta_1 lnS(q_c) + \beta_2 lnA + \beta_3 lnT, \tag{4}$$

where $\beta_1 = \beta_0\kappa_1$, $\beta_2 = \beta_0\kappa_2$, $\beta_3 = \beta_0\kappa_3$.

When $e = 0$: The migrant parent sends his child back to the rural hometown where the child receives free compulsory education.

Assuming the maximization problem has an interior solution,[13] the optimal time spent with the child in rural hometown is

$$t^*_{e=0} = \zeta \frac{E(w_c|X)}{E(w_c|X) - E(w_r|X)} T = \zeta \frac{1}{1 - E(w_r|X)/E(w_c|X)} T, \tag{5}$$

where $\zeta = \frac{\beta_3}{1+\beta_3}$. The utility in the scenario $e = 0$ is

$$U_{e=0} = ln[(1-\zeta)E(w_c|X)T] + \beta_0 ln\kappa_0 + \beta_1 lnS(q_r) + \beta_2 lnA + \beta_3 ln[\zeta \frac{E(w_c|X)}{E(w_c|X) - E(w_r|X)} T]. \tag{6}$$

Second, a migrant parent decides whether to send his child back to the rural

[13] The existence of the interior solution requires that $\frac{\beta_3}{1+\beta_3} \frac{E(w_c|X)}{E(w_c|X)-E(w_r|X)} < 1$.

hometown or to keep his child in the city, using the following decision rule

$$e = \begin{cases} 0, & \text{if } U_{e=1} < U_{e=0}; \\ 1, & \text{if } U_{e=1} \geq U_{e=0}. \end{cases}$$

We consider the difference in utilities,

$$U_{e=1} - U_{e=0} = ln[(1 - \frac{\theta(X, d_c)}{E(w_c|X)T}) \times (\frac{S(q_c)}{S(q_r)})^{\beta_1} \times (1 - \frac{E(w_r|X)}{E(w_c|X)})^{\beta_3} \times \frac{1}{(1-\zeta)\zeta^{\beta_3}}], \quad (7)$$

where $\frac{\theta(X,d_c)}{E(w_c|X)T}$, the ratio of education costs over the potential earnings in the city, represents the relative financial burden of enrolling the child in an urban school.

The above simple conceptual framework offers one important insight. The likelihood of a child being left behind in the rural hometown is jointly determined by both Equations (5) and (7). Higher school enrollment cost $\theta(X, d_c)$ induced by more intensive policy pressure d_c elevates the child's risk of being left behind in the rural hometown. Recall that $\frac{\partial\theta(X,d_c)}{\partial X} < 0$, this risk is expected to be more pronounced among families with a lower social-economic status.

4 Data and Econometric Model

4.1 Data and Measurement

The primary data used in the analysis is from the Chinese Migrant Dynamic Monitoring Survey (2011-2016, CMDMS) conducted by the China National Health and Family Planning Commission. The survey covers 31 provinces in China, and the sample size ranges from 128,000 in 2011 to 158,460 in 2016. So far, this is the largest nationally representative data set for migrants in China. It contains detailed information about demographic characteristics, social-economic status, migration history, and family living arrangements for all household members, including those who are currently not living in the surveyed city (Lyu and Chen, 2019). We complement this

dataset with a rich set of covariates at the city-year level, including city income level, labor market environment, and public service supply measures obtained from the China City Year Book and the population data in the city district area from the China City Construction Year Book.

We restrict our sample to *mega cities* as they are subjected to population control policy. Following the "National New-Type Urbanization Planning (2014-2020)," the *mega cities* are defined as those with a population of no less than five million in the central district in 2013, the last year before the policy. The resulted sample includes 11 cities: Beijing, Shanghai, Guangzhou, Shenzhen, Dongguan, Nanjing, Tianjin, Wuhan, Zhenzhou, Chongqin, and Shenyang.

Responding to the call by the central government, each *mega city* announced its targeted population goal by 2020 and had high flexibility in design its own initiatives to control the migrant inflow so that the total city population does not surpass the targeted goal by 2020. We construct a policy pressure index for each city to measure the local government's determination to implement a stringent policy. The index is the ratio of the city population in 2013 over the target population in 2020 in the city planning, quantifying how stringent the policy would be implemented. The higher the policy index, the larger pressure the local government bears to control the city population. We collect the documented population goals in 2020 from city official documents.[14] In Table 1, we list the population goal by 2020 and city population in 2013 for each city. Dongguan, Shanghai, Beijing are the top three cities with the highest policy pressure, and Shenyang and Shenzhen are ranked the bottom two with the least policy pressure.

To explore the effect on parent-child separation among children at the compulsory-education age, we focus on children who are between 6 and 15 years old and from

[14]In official documents, all cities except Shenyang target at city population, while Shenyang sets the goal at the city central district population. In columns (3) and (4), we present population goal and population in 2013 at the city level for all cities except Shenyang; for Shenyang, we use population goal and population in 2013 both at the district level.

migrant families with rural *hukou*.[15] Our primary outcome variable is a binary variable indicating a child's left-behind status with a value of one if the child does not live in the same city with its interviewed migrant parent, otherwise zero.[16] Columns (1) and (2) in Table 2 summarize the migrant children sample. On average, 39 percent of the migrant children are left behind in the rural hometowns, 79 percent (0.309/0.39) of whom are left behind alone without any parent's companion. The migrant children are, on average, 10 years old, and more than 56 percent are male.

4.2 Empirical Model

As the local administrations of *mega cities* started the population control policy in 2014, we define 2011, 2012, and 2013 as the pre-policy period and 2014, 2015, and 2016 as the post-policy period. Our difference-in-differences (DD) model estimation exploits the variations in the policy pressure across cities, as specified below,

$$y_{ict} = \beta_0 + \gamma_1 PolicyPressure_c \times Post_t + \beta_1 X_{ict} + \beta_2 Z_{ct} + \theta_t + \lambda_c + \epsilon_{ict}. \quad (8)$$

In our baseline analysis on parent-child separation, y_{ict} denotes a binary variable indicating whether a child i is left behind by his parents who work in the city c in year t. The primary variable of interest is the interaction term, $PolicyPressure_c \times Post_t$. Its coefficient γ_1 quantifies the impact of one percentage point increase in the policy pressure. X_{ict} is a set of child's or parents' characteristics, as reported in Table 2.

We also control for the province of origin of the migrant family to take into account

[15]In China, as required by the Compulsory Education Law, children must attend primary school when they reach the age of six and complete nine years of education. Children between three to six years old are usually enrolled in preschools, but this is optional. Youth older than 16 years old are eligible to be employed. Here, we also restrict both parents to have rural *hukou*.

[16]This left-behind variable measures a child's exposure to the absence of at least one parent. We replicate our result by defining a left-behind child if he does not live in the same city with both migrant parents and obtain similar results. Besides, the left-behind children in this paper include those left behind either in their rural hometown or in other cities.

the economic development in the rural hometown.[17] In addition, we are concerned that some city-level environment factors correlated with the policy implementation may also affect parental choice. We further control for city characteristics in the previous year in Z_{ct}, which includes various city development measures, such as (log) GDP per capita in the city district, (log) area of the city district and pollution levels (PM10 intensity), labor market demand measures, such as (log) fixed investment and foreign direct investment per capita in the city, as well as city public service measures in the district areas, such as the general budget expenditure per capita and the share of education expenditure in the general budget. θ_t and γ_c are the year and city fixed effects, capturing the annual national-wide macroeconomic changes and city time-invariant features, respectively.

The standard errors are two-way clustered at city level and child age level to capture correlations within the city as well as correlations within a child age group in the analysis on parent-child separation.[18] Given that we have only 11 city clusters and ten child age clusters, for all the empirical exercises, we report wild-bootstrap cluster p-values and wild-bootstrap cluster 95 percent confidence intervals generated using boottest command (10,000 replications) following Roodman et al. (2019).

To ensure that the parallel trend assumption underlying the *DD* estimator is not violated, we also estimate an event study model as following,

$$y_{itc} = \beta_0 + \sum_{j \neq 2013} \gamma_j PolicyPressure_c * 1\{Year = j\} + \beta_1 X_{ict} + \beta_2 Z_{ct} + \theta_t + \lambda_c + \epsilon_{ict}, \tag{9}$$

where $1\{Year = j\}$ is an indicator variable for year j and the policy pressure index is interacted with each year indicator. The coeffiicent γ_j stands for the policy impact in

[17]The CMDMS data do not include information on cities of origin. To the best we can do, we control for the province of origin fixed effect.

[18]We use the multi-way clustering in the child analysis because the children within a particular age group may be in the exposure to a common shock from national education policy and are correlated. For example, children at age six are mandated to enroll in primary school.

the corresponding year j compared to the reference year of 2013. This model allows a more flexible investigation into any differential trends in our interested outcome across cities in the pre-policy years. The standard errors are addressed in the same way as in Model (8).

4.3 Test Compositional Change

We also perform empirical exercises examining the policy impact on the compositional changes of migrant families using a sample of migrant families whose head is at prime-age (20 to 55 years old) and has rural *hukou*. This sample includes both migrant families with children at the compulsory-education age and also those without. As shown in columns (3) and (4) in Table 2, it is not surprising that this sample of migrants are younger and less like to be married than the parents in children's sample as shown in column (1) and (2). About 69 percent of family heads have a middle school diploma or less; 56 percent of the families have at least one child under 15 years old and 34 percent have at least one child at the compulsory-education age.

We use the same empirical framework as Models (8) and (9). The outcome variable, y_{ict}, can be a binary variable on whether a migrant family obtains a degree no more than middle school or a dummy on whether a migrant family has at least one child at the compulsory-education age (6 to 15 years old). The standard errors are clustered at the city level to capture the correlation within the city. We report wild-bootstrap cluster p-values and wild-bootstrap cluster 95 percent confidence intervals.

5 Impact on Parent-Child Separation

We first examine how the education-based migration control policy affects migrant parents' decisions on children's arrangements. In particular, we adapt the cross-city difference-in-differences design and investigate the policy impact on the probability of a child being left behind.

5.1 Baseline Results

Table 3 summarizes our main findings on parent-child separation from Model (8). In column (1), the point estimate is 0.430 and significant at the 5 percent significance level, showing that one percentage point increase in the policy pressure is about to cause 0.43 percentage points rise in the probability of children being left behind. Given that the proportion of left-behind children is 0.384 prior to the policy among the *mega cities*, this estimate suggests that if a city lowers its target population goal in 2020 by 0.8 percent, the proportion of children from the migrant families in the city being left behind will increase by 1.12 percent.[19] Figure 1, plotting the coefficients of the interactions between policy pressure index and each year indicators in Model (9), provides graphic evidence illustrating that the proportions of left-behind children across *mega cities* evolved in parallel before policy: the point estimates for 2011-2012 are centered around zero and statistically insignificant. The estimated policy effect increases to 0.4 noticeably in 2014 after the policy initiation, and the magnitude persists over time.

Aware of the prevalent "son preference" in rural China, we examine the policy impact on boys and girls separately in columns (2) and (3). It is shown that both boys and girls are strongly affected by the policy; however, the magnitude of the influence on girls is more than twofold that on boys. With one percentage point increase in the policy pressure, the probability of being left behind for girls rises by 0.6 percentage points, compared to an increase of 0.29 percentage points for boys. Figures (a) and (b) in Figure 2 also echo the notable differences in the policy effect between boys and girls. After 2014, the policy effect on boys stabilizes and centers around 0.4, whereas the policy effect on girls is much stronger, gradually increasing through 2016. Note that for both genders, we find no indications of the violation of the parallel trend assumption. The finding of "son preference" in the children's arrangement decision raises our concern

[19] The mean of policy pressure index over the 11 *mega* cities is 0.836. Given that the population in 2013 does not change, one percentage point increase in the policy pressure index is equivalent to a 0.8 ($= 0.01 * 0.836$) percent change in the policy pressure index. Besides, the average proportion of left-behind children is 0.384 before policy. Then 0.43 percentage point change in the probability of being left behind is equivalent to 1.12 ($= 0.43/38.4$) percent change.

that, as girls are more likely to be sent back to rural hometowns where the quality of education is comparatively lower than that in the cities, and to grow up in the absence of parents, the gender gap in China could be exacerbating in the near future, which is an issue worthy of attention.

We further explore two different definitions of left-behind status. We concentrate on whether a child is left behind alone, that is, in the absence of both parents, in column (4) and whether a child is left behind with one parent in column (5). In both scenarios, the reference group is children staying in the city with their parents. It appears that upon the policy initiation, children are more likely to be sent back and live without either parent's companion than to be left behind with one parent. The estimated effect on children being left behind alone is nearly identical to the baseline estimator in column (1), whereas the effect on children being left behind with one parent is less than half of that in column (1) and statistically insignificant. The visualizations of the event study in figures (c) and (d) in Figure 2 are in line with our point estimates in the table. This is not surprising as we note that in Table 2, approximately 79 percent of the children left behind are those left behind alone.

5.1.1 Heterogeneous Impacts among Migrant Households

It is worth noting that the migration control policy is skill-biased, as *mega cities* intend to upgrade human capital in the migrant labor force by restricting the inflow of low-skilled workers. Thereby, we would expect to see that children from families of low socio-economic status are hit the most. In light of this, we compare the DD estimators from migrant families with different skill levels: *first*, parents whose educational attainment is no more than middle school *versus* those with at least high school education; *second*, families with an annual income less than the city average income *versus* those above. The results are reported in Table 4.

We find that the impact of the migration control policy is mainly evident among vulnerable migrant families. Specifically, children from less-educated or low-income families are more likely than their counterparts from better social-economic status families

to be opted out of the education system in the cities, be sent back, and be left behind in the rural hometowns. The coefficient from the less-educated families is 0.527, more than three times as large as that from the more educated families. Likewise, the coefficient from high-income families is less than one-third of the effect from the poor, let alone its large standard error. The paths of the coefficients in Figure 3 provide graphical evidence supporting the heterogeneous policy effects across families and meanwhile give minimal indications of violations against the parallel trend prior to the policy. We also notice a sizable jump in the policy effect in 2016 for the high-educated families. We suspect this is because local government attempts to implement the policy in a more restrictive way as the policy deadline is approaching.

5.1.2 Heterogeneous Impacts Across Children at Different Ages

To ensure that our estimates of the policy impact do not take up other macroeconomic shocks, we take advantage of the fact that the population control policy elevated the enrollment standard for primary and middle schools the most. The estimated effect above, if truly reflecting the educational initiatives, should show up primarily for children at compulsory-education age and less prominently for children beyond the compulsory education period.[20]

We examine the policy impact among children in four mutual-exclusive education periods: pre-school (2 to 5 years old), primary school (6 to 12 years old, compulsory education stage), middle school (13 to 15 years old, compulsory education stage), and beyond middle school (16 to 25 years old). The results are summarized in Table 5 and Figure 4. The point estimators in columns (2) and (3) reveal statistically significant and large impact for children in the compulsory education period. The estimated impact is 0.412 for children in primary schools and 0.534 for children in middle schools, both statistically significant and comparable to the baseline results in terms of magni-

[20] As long as a child finishes his compulsory education in the ninth grade and reaches 16 years of age, he can choose to stay with his/her parents and work in a city rather than separating from his parents to enter high school in his home province. For preschool children under six, their parents can choose not to send them to kindergarten, as preschool education is not mandatory.

tude. The paths of the coefficients in figures (b) and (c) in Figure 4 also resemble the pattern in our baseline figure, showing positive and persistent policy impacts after the policy initiation. The coefficient in column (4) is neither economically noticeable nor statistically significant. Its magnitude is less than one-fourth of the baseline, rendering no indication of distinguishable policy impact among children above junior middle school. We note that the impact among children of pre-school age is as large as 0.264 though marginally insignificant. We propose two possible explanations for the sizable coefficients in column (1). First, it might be because that the local governments in cities like Shanghai and Beijing also mandated the closure of the private-owned kindergartens whose students are primarily from migrant households. Besides, parents would also likely send their children home one or two years earlier than primary school if they expect their children to not qualify for public education in the cities. Consequentially, children between four and five years old are also likely to be left behind.

5.1.3 Ruling Out Other Alternative Explanations

We have thus far offered our preferred interpretation that the increase in the probability of migrant parent-child separation in *mega cities* is a consequence of a migrant parent's strategic response to the education-based migration control policy. To be cautious, we also suspect that some other concurrent events along with the migration policy may confound the estimated policy impact. Specifically, the rapid developments in the rural areas and the increasing air pollution in *mega cities* could both contribute to a migrant parents' incentive to leave their child behind. Here, we propose several alternative explanations, but fail to find strong evidence supporting them. The results are presented in Table 6 and Figure 5.

The first potential alternative explanation is the air pollution in *mega cities*. Also starting from 2014, Chinese cities started to release an air pollution index to the public. Since then, the public has been more aware of the severe adverse effect of air pollution on health. Recent studies find marked evidence in support of pollution-driven migration decisions. Chen et al. (2017) finds that in China between 1996 and 2010, air pollution

in a given county contributed to a 50 percent reduction in the migration inflows and reduced the population by five percent as a consequence of an increase in net out-migration. Qin and Zhu (2018) finds that the expectation of worsening air quality the next day raises the web search index on "emigration". In our study context, an unpleasant scenario is that pollution levels could confound our baseline estimates if cities with high population control pressures also experience worsening environmental quality. However, if pollution is the concern, then we should note a more prominent impact for the well educated or high-income families,[21] and among young children who are thought to be more vulnerable to the adverse effects of air pollution. In contrast to our conjecture, we find that the increase in the probability of children being left behind is more pronounced for low-skilled parents (see Table 4). Again recall that the estimated influence is only considerable for children in compulsory schooling but not for children under the age of six, as shown in Table 5. Besides, our main specification also accounts for the city's average pollution level (PM10 intensity) in the previous year to capture the environmental quality in the city.

Another possibility is the increasing transportation convenience across cities. Migrant parents may find it less costly for them to visit their children in their rural hometowns. The most impressive transportation development in China is the high-speed railway (HSR) network. The HSR network has expanded rapidly since 2008 and significantly reduces the transportation time cost between destination cities and migrants' hometowns if they are connected in the HSR network.[22] We directly test this hypothesis by including dummy variables indicating whether or not there is a direct or indirect HSR connection between the destination cities and the capital city of the

[21]For instance, the findings of Chen et al. (2017) are primarily driven by well-educated people at the beginning of their careers.

[22]In 2008, the first HSR line, the Beijing-Tianjin inter-city HSR line, was launched. Up to December 2018, the total HSR millage in China reached 29,000 km, accounting for two-thirds of the total HSR mileage in the world. The newly-built HSR is designed to reach 350 km per hour. It is only a 4.5-hour HSR trip from Shanghai to Beijing which is 1,213 km apart. It costs approximately 0.066 USD per km.

province of origin.[23] Our main results remain consistent, as shown in column (1) in Table 6.

The third potential confounder is the high cost of living in *mega cities*. Housing prices have been rapidly increasing in the recent decade in *mega cities* (Fang et al., 2016, Glaeser et al., 2017). Consequently, rental prices have also witnessed a sizable increase for low-skilled migrants. Bringing children together to *mega cities* may cost more as migrants need to rent a larger place and may not wish to share accommodation with others. Therefore, the budget constraint could lead migrant parents to flow into the cities, leaving their children behind. We incorporate the annual average housing price in our model in column (2) to capture the growing trend in housing expenses. The result is effectively unchanged, and, if anything, the estimate becomes slightly smaller in absolute value relative to the baseline.

Another possibility is the fast growth of education supply in the origin cities. The development in the rural areas reduces the education gap between *mega cities* and the origin cities.[24] However, if this is the explanation driving our baseline findings, we should observe a continuously increasing trend in the probability of children left behind in *mega* cities, other than a "kink jump" after 2014 in Figure 1. Further, to rule out this possibility, we control for the dynamic changes in the province of origin by adding the origin province linear year trend in column (3), and our result remains. In column (4), we incorporate all the factors mentioned above into one model, and the result is also consistent with our baseline.

To summarize, the heterogeneity in the impacts across different skill groups is

[23]The data does not contain the origin city information. We include three dummies to capture the status of the HSR connection in terms of whether or not the city has a high-speed railway station in a particular year: first, both the destination city and the capital city of origin province are connected; second, the destination city is connected, but the capital city of origin province is not; and third, the destination city is not connected, but the capital city of origin province is connected. The reference group is neither the destination city nor the capital city of origin province is connected.

[24]As students are required to take the college entrance examination back in the domicile of their hukou registration, migrant students have to come back to their rural hometown to attend high school. Some parents may find it is better to send their children back early if the education quality in rural increases.

resonant with the skills-biased feature of this migration control policy. The most pronounced impact on children at the compulsory education stage reassures the educational pathway underlying the estimated policy impact. We also attempt to rule out other possible interpretations that are not backed up by any empirical evidence. Analysis all above suggests that the education-based migration control policy primarily drives the parent-child separation within migrant families.

5.2 Suggestive Evidence Using Education-Access Barrier Metric

In this section, we construct an index to measure the difficulties of enrolling a migrant child in public schools in each *mega city* and evaluate its impact on parent-child separation as supplementary evidence.

The education-access barrier index is calculated based on the enrollment requirements mandated by local city administrations. Each city is entitled with its own jurisdiction to specify a series of criteria determining the eligibility of a migrant child to attend a public school. These criteria include, but are not limited to, a legal residency certificate in the city, time length that parents have stayed in the city, ownership of any purchased dwelling, and payment of social security insurance. We collect these data for seven cities (Beijing, Shanghai, Nanjing, Tianjin, Guangzhou, Wuhan, and Zhenzhou) in 2016.[25] The education-access barrier index for each city in 2016 is calculated and then standardized using the mean and standard deviation of the seven scores. As shown in Figure 6, a city bearing a more considerable pressure in controlling the population also tends to have a higher education-access barrier index.

[25]Generally speaking, there are two schemas to determine if a migrant child is eligible to enroll in the city public education system. One is called "documents-based" schema, and the other is called "score-based." Beijing, Shanghai, Nanjing, Tianjin, Guangzhou, Wuhan, and Zhenzhou adopt the document-based schema, requiring parents to submit certain required legal documents to apply for a seat. Chongqing, Dongguan, and Shenzhen take up the "score-based" schema, which scores migrant parents based upon a list of items covering education background, tax contribution, housing status, and occupation characteristics. As stated in Han et al. (2020), the difficulty scores in these two systems are not comparable. Hence we only include cities using the document-based schema.

$$y_{ic} = \beta_0 + \gamma_1 Score_c + \beta_1 X_{ic} + \lambda_c + \epsilon_{ic}, \tag{10}$$

We perform Model (10) using migrant children samples from the above seven cities in 2016. Results are organized in Table 7. In column (1), our analysis focuses on migrant children between six and 15 years old. It shows that one standard deviation increase in the barrier to education access significantly increases the likelihood of migrant children being left behind. In columns (2) and (3), placebo tests conducted among migrant children at pre- or post-compulsory education age do not render significant impacts. To be cautious, we acknowledge that findings from Table 7 may not ideally offer a plausible causal interpretation as we only include seven cities that uses the document-based schema. We view the results as suggestive evidence complementing our primary conclusion by directly investigating the relationship between access to education and the parent-child separation.

6 Test Compositional Change in Migrant Families

Compositional changes in migrant families in cities over time may threaten the validity of our DD estimates. In what follows, we assess how the migration policy affects the composition of migrant families in cities from three angles. we first look at the education levels. Less-educated migrant labor is less likely to meet the school enrollment requirements in the public education system in the city. If they prioritize their children's education over job opportunities in the city, they may leave for other cities or hometowns. Consequentially, we will witness an reduction in share of less-educated migrant families. Besides, the population policy is less binding for families without children at the compulsory-education age. Thereby, we further investigate whether parents without children at the compulsory-education age suffer less risk of leaving the city than those with. Lastly, we explore the policy effects on spousal living arrangement. In this section, we mainly focus on the sample of migrant families (heads).

6.1 Compositional Change in Education Levels

Table 8 presents the results with an indicator for whether a migrant's education attainment is lower than middle school as the dependent variable. The estimates across different sample specifications do not speak to any apparent differences driven by the policy.

In column (1) of Panel A, the coefficient is 0.196 without reaching standard statistical significance, showing that the policy does not impose any remarkable influence on the education structure of the migrant labor force. Note that families with children at the compulsory-education age are exposed to the education-based migration policy the most. Thereby, we further narrow our focus on the adult migrants with at least one child at compulsory education age in columns (2). The estimates are slightly smaller but still render no indications of a distinct policy effect. In column (3), we further exclude government officials as they are less likely to be bound by the policy, and the result remains similar. In column (4), we perform a placebo test in a sample of adults from families without compulsory-education-aged children, which assuringly confirms our findings from the first three columns.

Figure 7 exhibits the coefficients of the interactions between the policy pressure index and the indicator for each year from the event study analysis corresponding to each column in panel A in Table 8. The figures allow us to visualize the development of the policy influence over time. For instance, in figure (a), the coefficients prior to the policy initiation are not statistically different from the reference in 2013, which validates the parallel pre-trend assumption. The estimated coefficients in the post-policy years are not distinguished from zero. Consistent with our results in Table 8, the path of the coefficients in each figure does not offer any notable support for policy influences on the education structure of the migrants in cities.

There are salient variations in the educational attainment across cohorts due to the introduction of the compulsory education law in China. Those born after 1980 were mandated to finish a nine-year education by law; therefore, the majority of them are

middle school graduates. However, those born before 1980, on the other hand, have a lower level of education. To rule out the concern that this structural difference in the educational attainment between those two cohorts may mix the policy effect, we replicate our analysis in panel B, restricting our sample to those born after 1980. The estimates are in line with those in panel A.

6.2 Compositional Change in Family Structure

A natural question to ask is whether the education-based migration control policy phases out migrant families that have at least one child at the compulsory-education age, as those families are hit the most. In Table 9, we first examine the changes in the proportion of families with the presence of at least one compulsory-education-aged child among the married couple before and after the policy initiation in panel A. In column (1), the coefficient is -0.045 but not significant, implying a negligible reduction in the proportion of families with at least one compulsory-education-aged child owing to the policy. Aware that less-educated families may find it more demanding for them to meet the school enrollment requirements, we further restrict our analysis to families in which the education level of the family head is no more than middle school in column (2). We see negligible policy influences as well.

It is possible that families with younger children may expect that they cannot meet the requirement and leave the city before their children reach the age of six. To check this, we test whether the policy affects the proportion of migrant families with a child under 15 years old in panel B. None of the estimations can be distinguished from zero. In Figure 8, we visualize the estimators from the event study for each column separately. We do not see a notable sign for the existence of pre-trend, nor any distinguishable phase-out effects of the migration policy on families with at least one child after its implementation.

6.3 Change in Spousal Living Arrangement

It is still not clear how labor specification decision is made between spouses. One possible labor specification could be that one spouse stays in the city and the other leaves the city with their child when the child is not eligible for public school enrollment in the city. To investigate the effect on spousal separation, we create an indicator taking the value of one if spouses are not living together in the city, and zero otherwise. The results are displayed in Table 10 and Figure 9.

In column (1), the estimate from all spouses is 0.014, with its 95 percent confidence interval covering zero, which provides little evidence to support spousal separation. We restrict our sample to spouses with at least one child at the compulsory-education age in column (2) and spouses that have a diploma no more than middle school in column (3). The point estimates in columns (2) and (3) increase, but both of them are not statistically significant. In column (4), we further restrict to a sample of spouses with no more than middle school education and with at least one compulsory-education-aged child. The point estimate increases to 0.1 but remains insignificant. Recall that in Table 3, there is no indication that the share of children left behind with one spouse increases considerably after the policy; it is not surprising here to see a minimal evidence for the separation between spouses as a response to the education-based policy change.

7 Conclusion

This paper evaluates a newly implemented migration control policy in China among cities with more than five million population in 2013. Those cities, known as *mega* cities, have implemented a series of skill-biased initiatives to control the inflow of low-skilled migrant population, including imposing more stringent school enrollment requirements to limit migrant children's access to public schools.

We develop a theoretical framework to illustrate the policy consequences on the labor market and family disengagement. Our model shows that tightening school en-

rollments may not effectively control the inflow but may lead to an unintended adverse consequence: children are likely to be separated from migrant parents and left behind in rural hometowns.

We also build a difference-in-differences model exploiting variations in the policy stringency across cities. Our empirical findings suggest that migrant children are more likely to be left behind if their parents work in cities with higher population control pressure. This impact is more pronounced for families with lower socioeconomic status and for children at the compulsory-education stage.

Whether or not a migration control policy is a good means of urbanization is beyond the scope of this paper. However, our empirical results highlight the concern that the increasing parent-child separation driven by this migration control policy may impede the human capital accumulation for the country's sustainable development and contradicts the central government's goal to decrease the total population of left-behind children.

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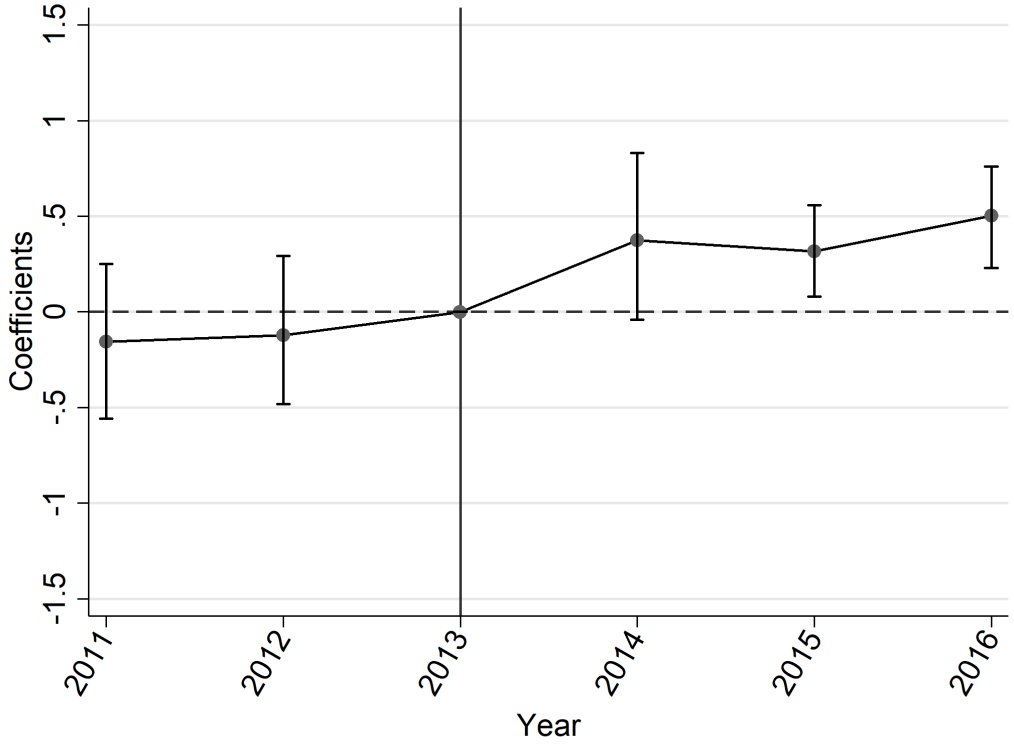


Figure 1: Impact of Migration Control Policy on Children Being Left Behind

Note: The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. We report wild bootstrap cluster 95 percent confidence intervals generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multi-way clustering: city level (11 clusters) and child age level (10 clusters).

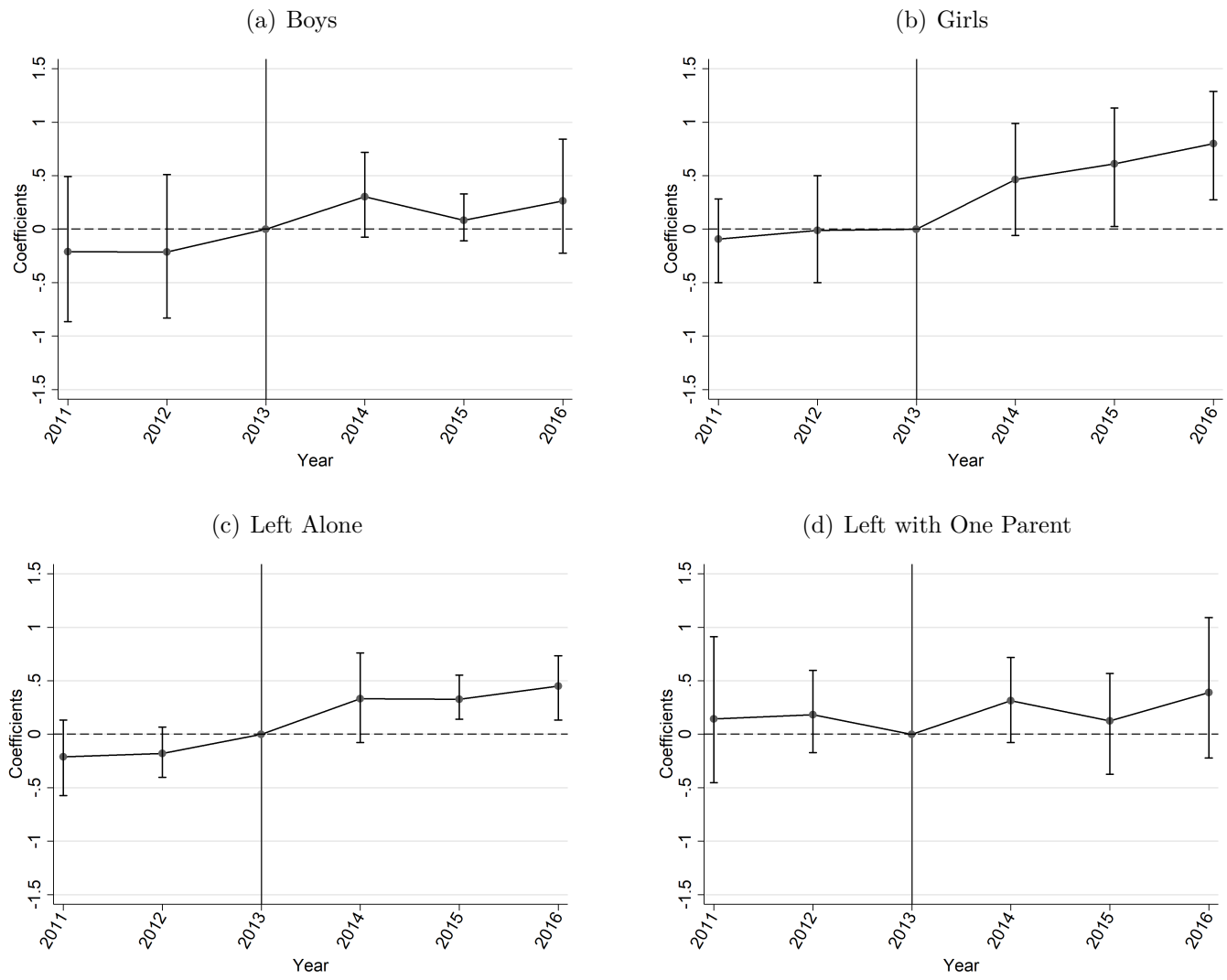


Figure 2: Impact on Children Being Left Behind: By Child Gender or Using Alternative Left Behind Measures

Note: The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. (a) and (b) present results from boys sample and girls sample, respectively. (c) and (d) present results using alternative left behind measures and the baseline sample. We report wild bootstrap cluster 95 percent confidence intervals generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multi-way clustering: city level (11 clusters) and child age level (10 clusters).

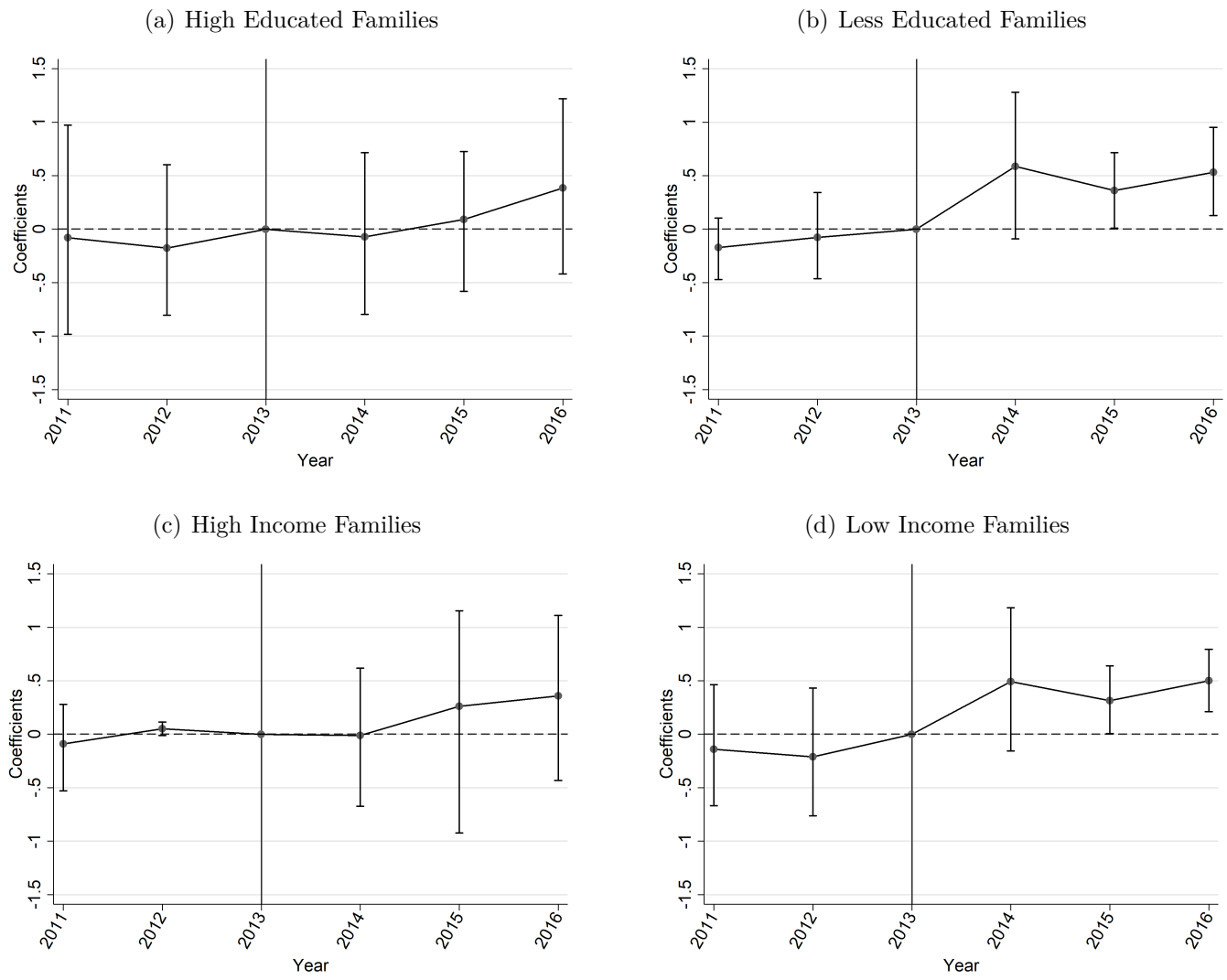


Figure 3: Impact on Children Being Left Behind: By Family Social Status

Note: The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. Sample in (a) and (b): whether the education of both parents is more than middle school; in (c) and (d): whether family income higher than average migrant family income in the city that year. We report wild bootstrap cluster 95 percent confidence intervals generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multi-way clustering: city level (11 clusters) and child age level (10 clusters).

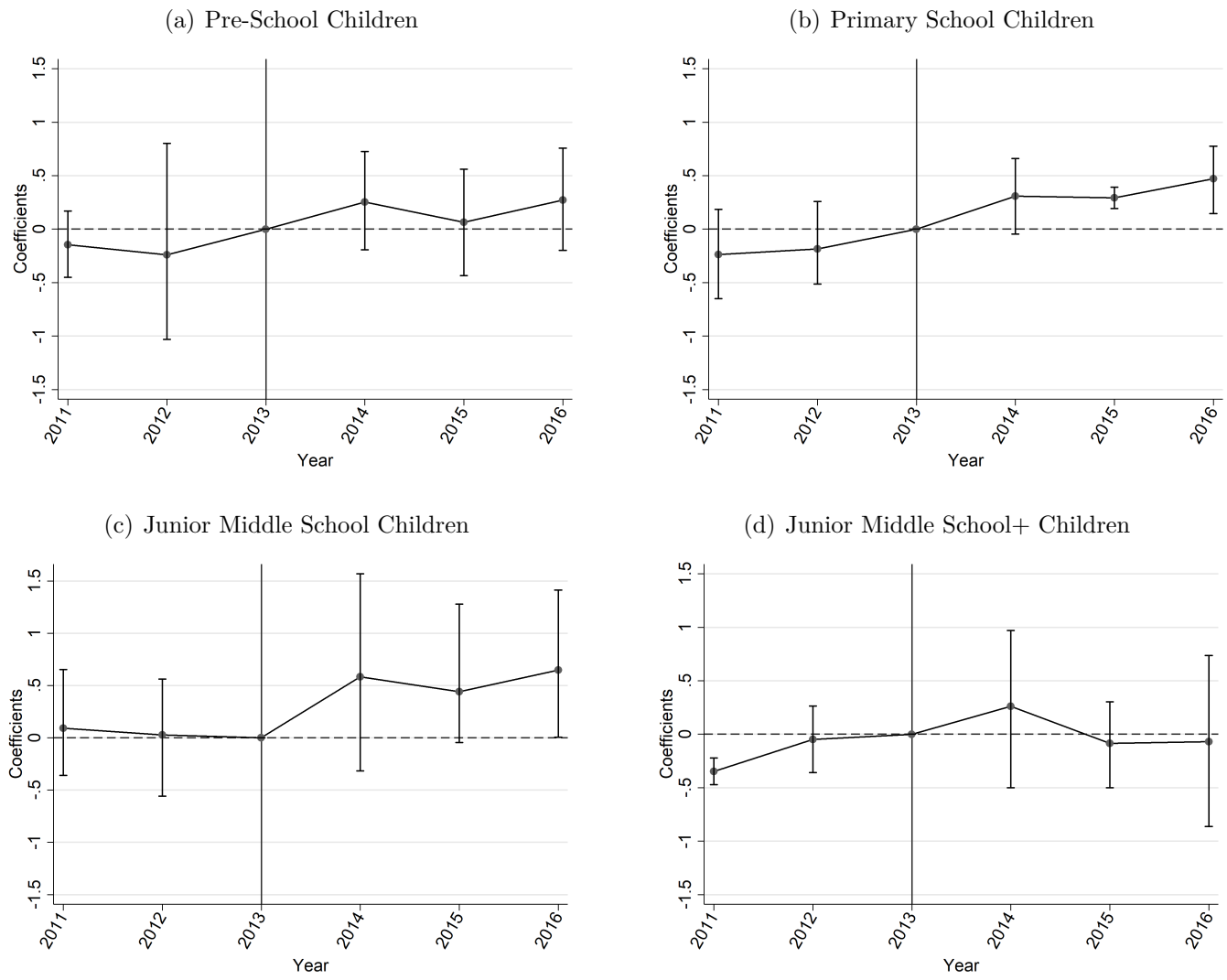


Figure 4: Impact on Children Being Left Behind: By Child Age

Note: The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. The education stage are classified as follows: 2-5 years old for pre-school period, 6-12 years old for primary school period, 13-15 years old for junior middle school period and 16-25 years old for junior middle school and above period.We report wild bootstrap cluster 95 percent confidence intervals generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multi-way clustering: city level (11 clusters) and child age level (10 clusters).

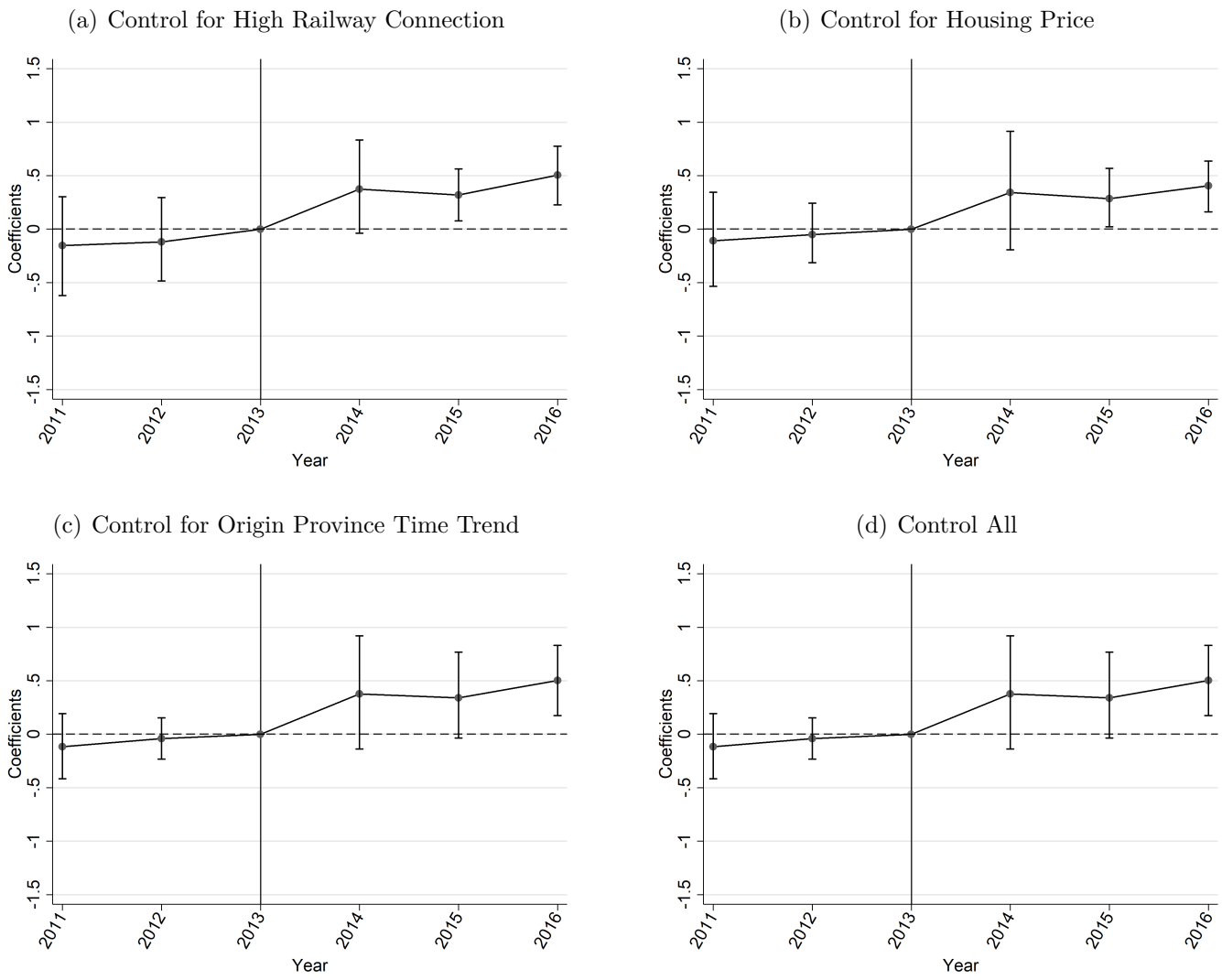


Figure 5: Impact on Children Being Left Behind: Channel Analysis

The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. We report wild bootstrap cluster 95 percent confidence intervals generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multi-way clustering: city level (11 clusters) and child age level (10 clusters).

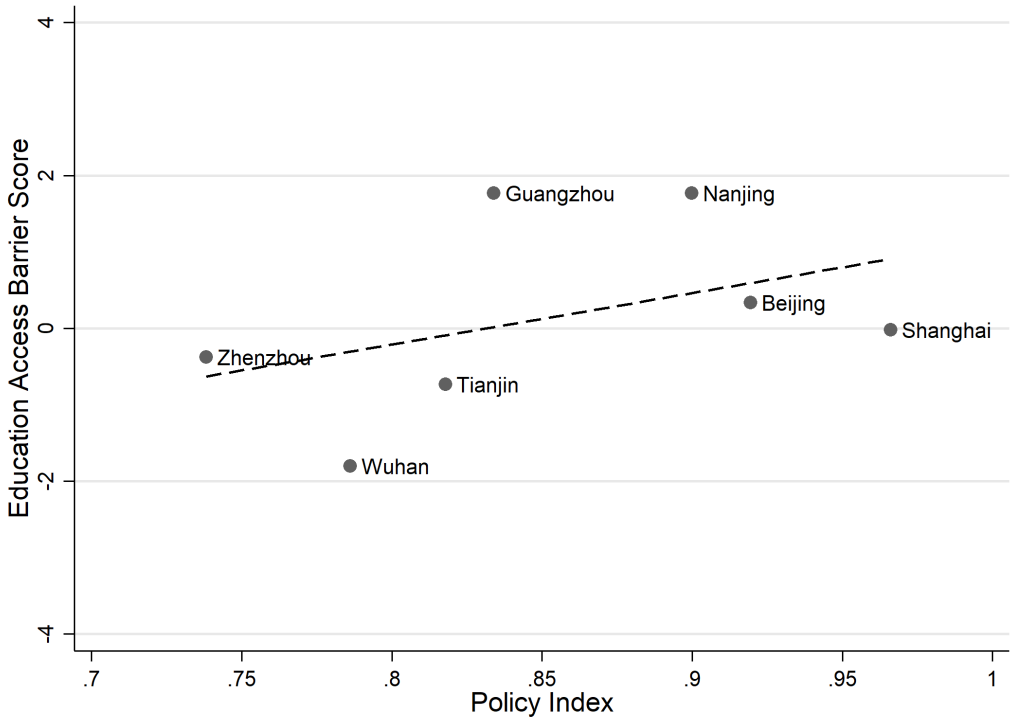


Figure 6: Education-Access Barrier Score and Policy Pressure Index

Note: Seven cities are in our analysis: Shanghai, Beijing, Nanjing, Tianjin, Guangzhou, Wuhan and Zhenzhou. We collect and calculate the education access barrier score in 2016 for each city. The dash line fits the points.

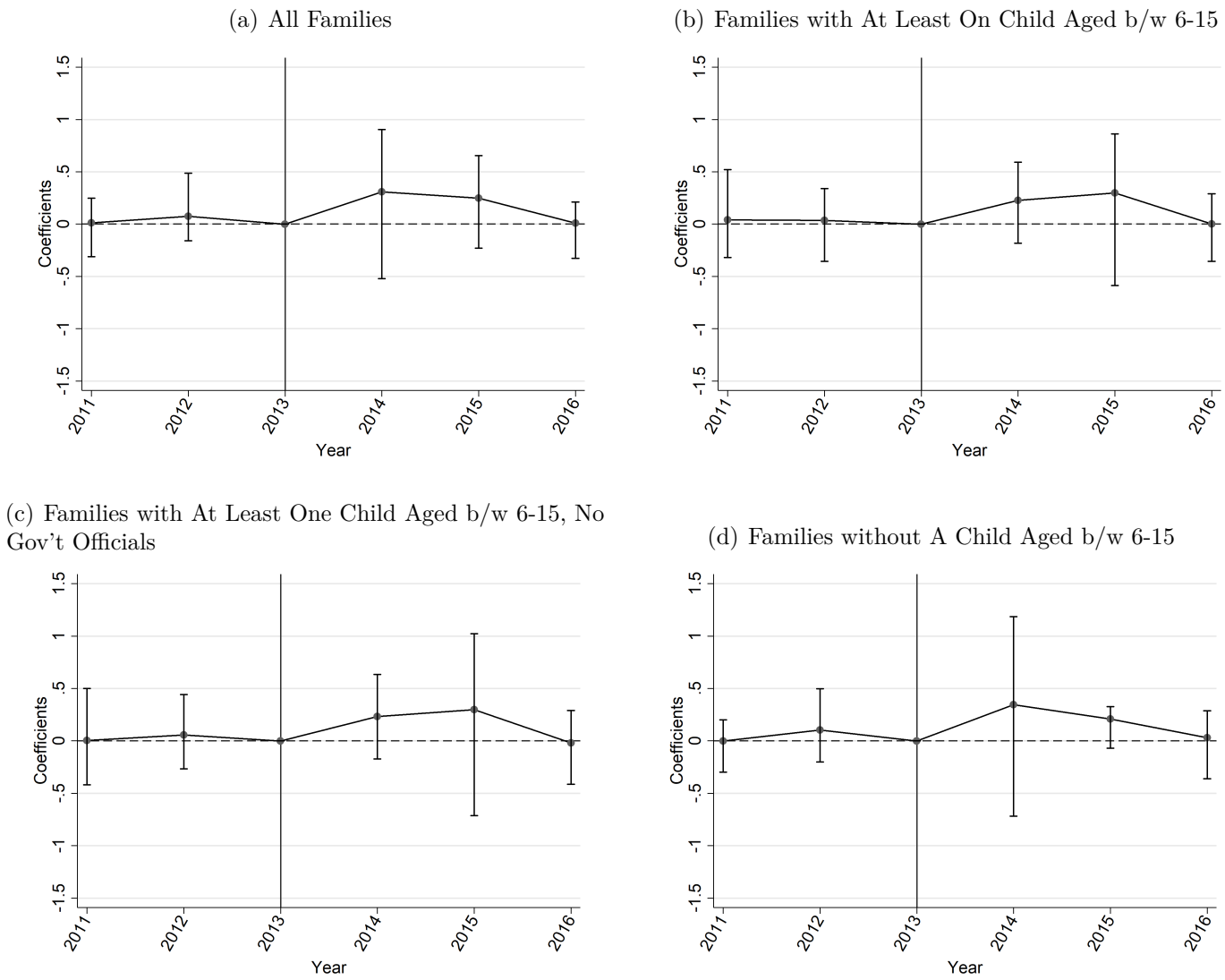
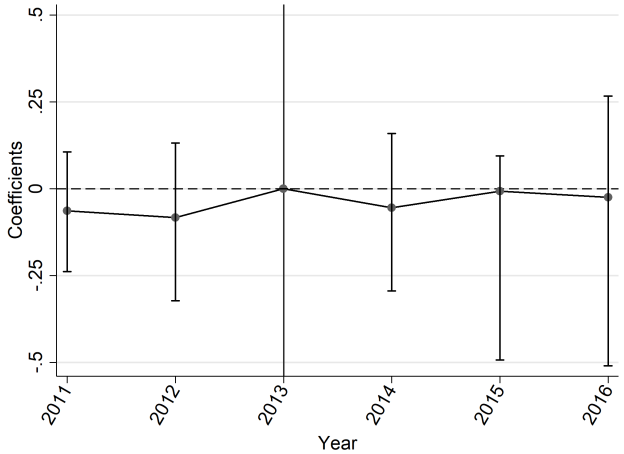


Figure 7: Test Compositional Change in Migrant Families: Education Composition

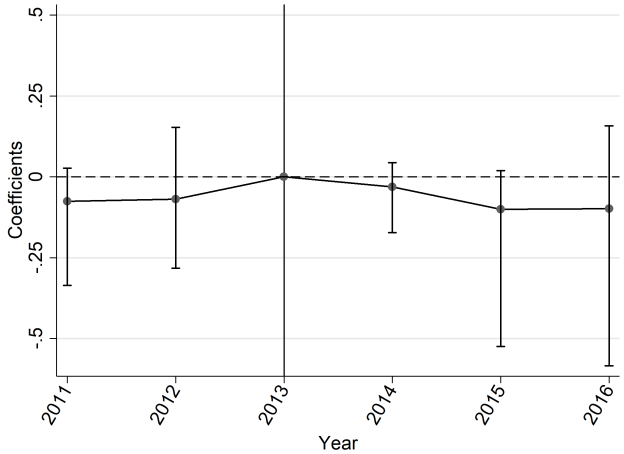
Note: We report wild bootstrap cluster 95 percent confidence intervals generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at city level (11 clusters).

Panel A, Outcome: Families with A Child Aged b/w 6-15

(a) Education ≤ Middle School

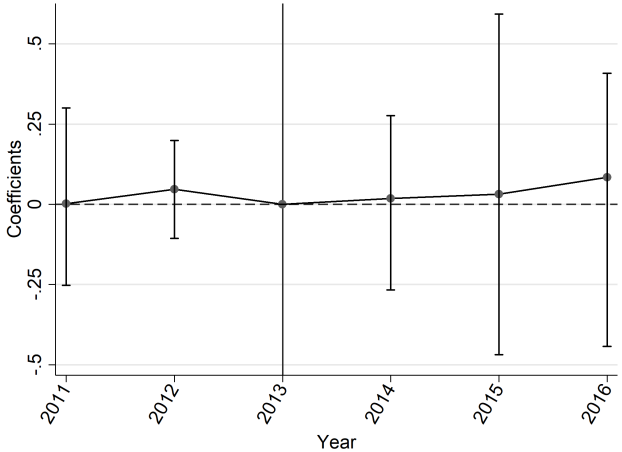


(b) Education ≤ High School



Panel B, Outcome: Families with A Child Aged b/w 6-15

(c) Education ≤ Middle School



(d) Education ≤ High School

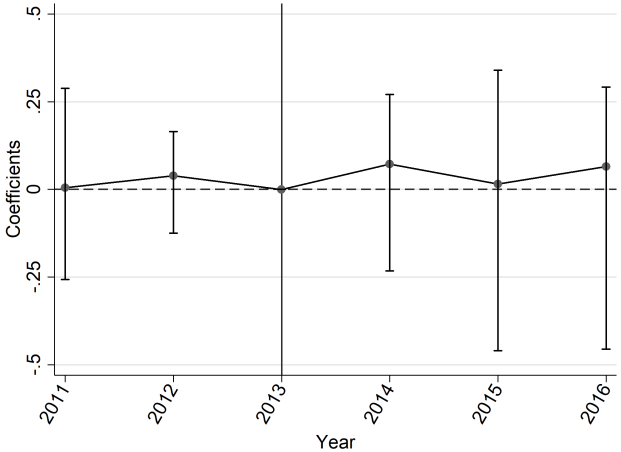


Figure 8: Test Compositional Change in Migrant Families: Family Structure Composition

Note: We report wild bootstrap cluster 95 percent confidence intervals generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at city level (11 clusters).

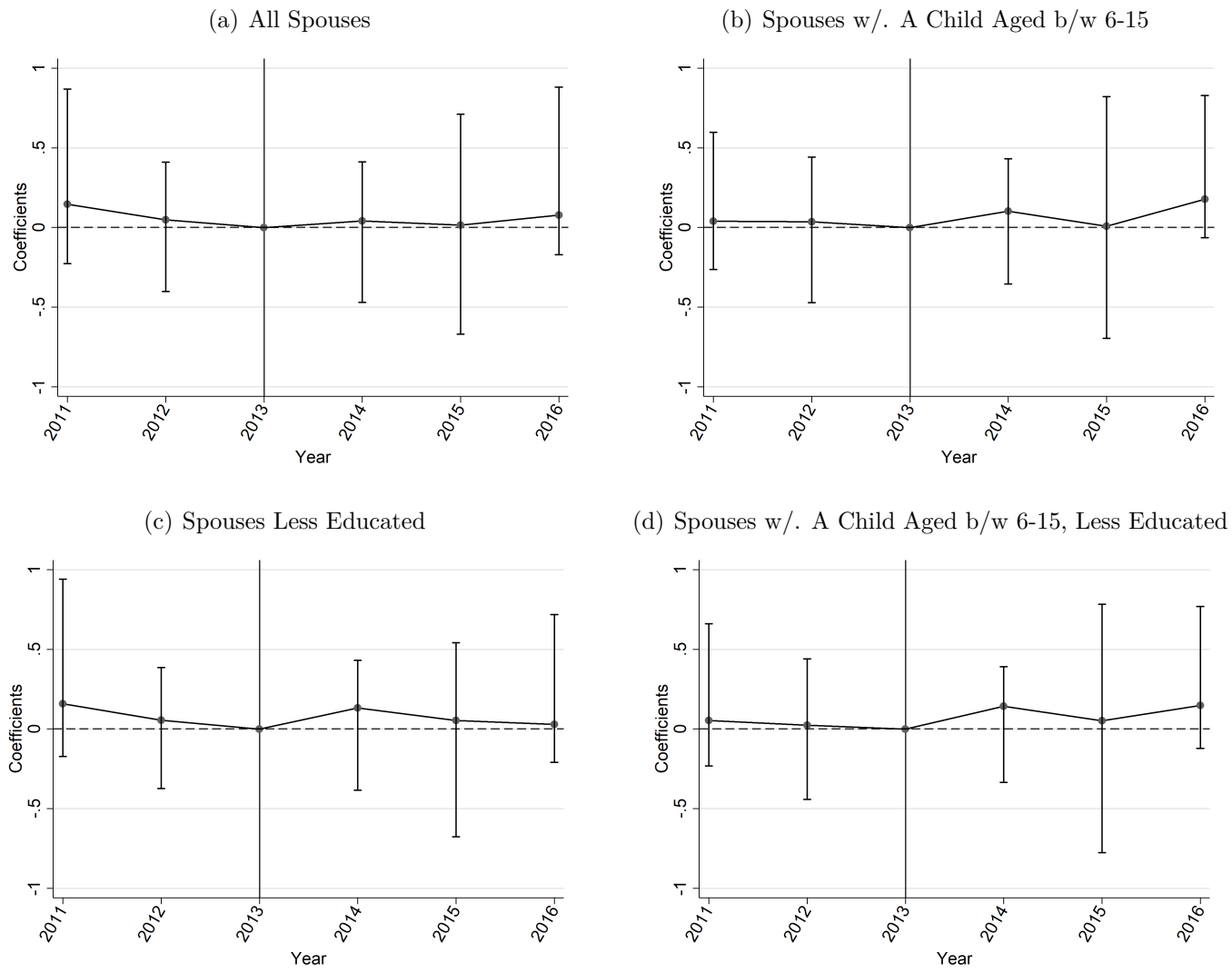


Figure 9: Impact of Migration Control Policy on Spousal Separation

Note: Spouses separation is an indicator taking value of 1 if spouses are not living together in the city, otherwise zero. We report wild bootstrap cluster 95 percent confidence intervals generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at city level (11 clusters).

Table 1: City Migration Control Policy

City	Policy Pressure Index	Population Goal in 2020	Population in 2013
(1)	(2)	(3)	(4)
Dongguan	0.978	850	831.66
Shanghai	0.966	2500	2415.15
Beijing	0.919	2300	2114.8
Nanjing	0.900	910	818.78
Guangzhou	0.834	1550	1292.68
Chongqing	0.821	3400	2790
Tianjin	0.818	1800	1472.21
Wuhan	0.786	1300	1022
Zhengzhou	0.738	1245	919.1
Shenyang	0.724	725	524.6
Shenzhen	0.718	1480	1062.89

Notes: The policy index is the ratio of population in 2013 over population goal in 2020 for each city. For Shenyang, the city migration control policy targets at the city district population, therefore, the district population in 2013 is reported for shenyang. For other cities, the city population control policy target at the city population, and we report the city population in 2013.

Table 2: Summary Statistics: Migrant Children and Families Sample

	Migrant Children Sample		Migrant Families Sample	
	Mean	Std. Dev	Mean	Std. Dev
	(1)	(2)	(3)	(4)
Children-Related Variables				
Child Being Left Behind	0.390	0.488	-	-
Child Being Left Behind Alone	0.309	0.462	-	-
Child Being Left Behind w/ One Parent	0.080	0.271	-	-
Male Child	0.568	0.495	-	-
Child Age	10.058	2.618	-	-
Family-Related Variables				
One-Child Family	0.353	0.478	-	-
Two-Children Family	0.548	0.498	-	-
Families w/ Any Child (0-15 yrs)	-	-	0.557	0.497
Families w/ Any Child (6-15 yrs)	-	-	0.341	0.474
Family Head Age	36.599	4.909	33.881	8.544
Education:				
Not Go to School	0.012	0.111	0.012	0.108
Primary School	0.143	0.350	0.112	0.316
Middle School	0.653	0.476	0.571	0.495
High School	0.170	0.376	0.238	0.426
University and Above	0.022	0.145	0.067	0.250
Log of Family Income (Monthly)	8.617	0.544	8.495	0.582
Current Flow Duration	6.314	5.221	5.032	5.052
Married	0.988	0.108	0.783	0.412
Male	0.514	0.500	0.523	0.499
Unemployed	0.135	0.341	0.139	0.346
Officials or Technicians	0.053	0.225	0.073	0.261
Service Workers	0.568	0.495	0.538	0.499
Agriculture Workers	0.011	0.105	0.009	0.094
Manufacture Workers	0.233	0.423	0.241	0.428
Observations	67799		165395	

Notes: Mean and standard deviations reported.

Table 3: The Effect of Migration Control Policy on Children Left Behind

Dependent Var:	Left Behind			Left Alone	Left w/. One Parent
Sample:	All Children	Boys	Girls	All Children	All Children
	(1)	(2)	(3)	(4)	(5)
Policy Pressure × Post	0.430**	0.290**	0.607**	0.427***	0.188
	(0.014)	(0.018)	(0.016)	(0.002)	(0.192)
	[0.235, 0.625]	[0.110, 0.495]	[0.252, 0.951]	[0.363, 0.493]	[-0.242, 0.578]
N	67799	38509	29290	62314	46765
Child Characteristics	Yes	Yes	Yes	Yes	Yes
Parent Characteristics	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Destination City FE	Yes	Yes	Yes	Yes	Yes
Original Province FE	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. We report wild bootstrap cluster p-values in parentheses and wild bootstrap cluster 95 percent confidence intervals in square brackets, generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multiway clustering: city level (11 clusters) and child age level (10 clusters). *** significant at the 1 percent level; ** significant at the 5 percent level and * significant at the 10 percent level.

Table 4: The Effect of Migration Control Policy on Children Left Behind – By Family Social Status

Sample:	High Edu	Less Edu	High Inc	Low Inc
	(1)	(2)	(3)	(4)
Dependent Var: Children Left Behind				
Policy Pressure × Post	0.159	0.527**	0.152	0.501**
	(0.191)	(0.032)	(0.533)	(0.027)
	[-0.174, 0.497]	[0.118, 0.954]	[-0.703, 0.843]	[0.149, 0.889]
N	17286	50513	17394	50405
Child Characteristics	Yes	Yes	Yes	Yes
Parent Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Destination City FE	Yes	Yes	Yes	Yes
Original Province FE	Yes	Yes	Yes	Yes

Note: The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. Sample in columns (1) and (2): whether the education of both parents is more than middle school; in columns (3) and (4): whether family income higher than average migrant family income in the city that year.We report wild bootstrap cluster p-values in parentheses and wild bootstrap cluster 95 percent confidence intervals in square brackets, generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multiway clustering: city level (11 clusters) and child age level (10 clusters). *** significant at the 1 percent level; ** significant at the 5 percent level and * significant at the 10 percent level.

Table 5: The Effect of Migration Control Policy on Children Left Behind – By Children Age

Dependent Var: Children Left Behind				
Sample:	Pre-School	Primary School	Junior Middle School	Above Junior Middle School
	(1)	(2)	(3)	(4)
Policy Pressure × Post	0.264	0.412***	0.534**	0.098
	(0.101)	(0.008)	(0.045)	(0.481)
	[-0.123, 0.678]	[0.289, 0.517]	[0.0328, 1.317]	[-0.338, 0.465]
N	38490	49728	18071	47514
Child Characteristics	Yes	Yes	Yes	Yes
Parent Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Destination City FE	Yes	Yes	Yes	Yes
Original Province FE	Yes	Yes	Yes	Yes

Note: The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. The education stage are classified as follows: 2-5 years old for pre-school period, 6-12 years old for primary school period, 13-15 years old for junior middle school period and 16-25 years old for the above junior middle school period. We report wild bootstrap cluster p-values in parentheses and wild bootstrap cluster 95 percent confidence intervals in square brackets, generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multiway clustering: city level (11 clusters) and child age level (10 clusters). *** significant at the 1 percent level; ** significant at the 5 percent level and * significant at the 10 percent level.

Table 6: The Effect of Migration Control Policy on Children Left Behind – Channel Analysis

Dependent Var: Children Left Behind				
Control for:	HSR	House Price	Original Province Time Trend	HSR House Price Original Province Time Trend
	(1)	(2)	(3)	(4)
Policy Pressure × Post	0.431**	0.343**	0.487**	0.380**
	(0.015)	(0.027)	(0.014)	(0.030)
	[0.245, 0.628]	[0.0935, 0.609]	[0.272, 0.708]	[0.0804, 0.709]
N	67799	67799	67799	67799
Child Characteristics	Yes	Yes	Yes	Yes
Parent Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Destination City FE	Yes	Yes	Yes	Yes
Original Province FE	Yes	Yes	Yes	Yes

Note: The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. We report wild bootstrap cluster p-values in parentheses and wild bootstrap cluster 95 percent confidence intervals in square brackets, generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multiway clustering: city level (11 clusters) and child age level (10 clusters). *** significant at the 1 percent level; ** significant at the 5 percent level and * significant at the 10 percent level.

Table 7: The Effect of Migration Control Policy on Children Left Behind – Education Access Barrier Score

Dependent Var: Children Left Behind			
Sample:	Children (6-15)	Children (2-5)	Children (16-25)
	(1)	(2)	(3)
Education Access Barrier Score	0.483**	0.289	0.313
	(0.019)	(0.140)	(0.269)
	[0.203, 0.758]	[-0.468, 1.221]	[-0.518, 1.047]
N	7702	4183	4626
Child Characteristics	Yes	Yes	Yes
Parent Characteristics	Yes	Yes	Yes
Destination City FE	Yes	Yes	Yes
Original Province FE	Yes	Yes	Yes

Note: The dependent variable is a binary outcome, taking value of 1 if a child is left behind, otherwise 0. Seven cities are in our analysis: Shanghai, Beijing, Nanjing, Tianjing, Guangzhou, Wuhan and Zhenzhou. We collect and calculate the education access barrier score in 2016 for each city. We report wild cluster p-values in parentheses and wild bootstrap cluster 95 percent confidence intervals in square brackets, generated using boottest command (10000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at multiway clustering: the city level (7 clusters) and child age level (10 clusters).Robust standard errors are in parentheses. *** significant at the 1 percent level; ** significant at the 5 percent level and * significant at the 10 percent level.

Table 8: Test Compositional Change in Migrant Families: Education Composition

Outcome: Indicator for No More than Middle School				
Sample:	All	HH w/. A Child Aged b/w 6-15		HH w/o A
		All	Not Govn't Officials	Child Aged b/w 6-15
	(1)	(2)	(3)	(4)
Panel A: Sample - Family Heads (20 to 55 yrs)				
Policy Pressure × Post	0.196	0.186	0.178	0.196
	(0.233)	(0.196)	(0.209)	(0.243)
	[-0.199, 0.508]	[-0.197, 0.431]	[-0.177, 0.428]	[-0.244, 0.552]
N	165395	56449	53316	108946
Panel B: Sample - Family Heads Born After 1980				
Policy Pressure × Post	0.201	0.201	0.195	0.192
	(0.274)	(0.226)	(0.271)	(0.341)
	[-0.275, 0.577]	[-0.234, 0.508]	[-0.232, 0.539]	[-0.358, 0.648]
N	138038	50990	48111	87048
Personal Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Destination City FE	Yes	Yes	Yes	Yes
Original Province FE	Yes	Yes	Yes	Yes

Note: Sample consists of families whose head is at prime age (20 to 55 years old), and has rural hukou. We report wild bootstrap cluster p-values in parentheses and wild bootstrap cluster 95 percent confidence intervals in square brackets, generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at city level (11 clusters). *** significant at the 1 percent level; ** significant at the 5 percent level and * significant at the 10 percent level.

Table 9: Test Compositional Change in Migrant Families: Family Structure Composition

Sample:	All	HH ≤ Middle School		HH ≤ High School
		All	Not Govn't Officials	
	(1)	(2)	(3)	(4)
Panel A: Outcome - Indicator for Families with A Child Aged 6-15)				
Policy Pressure × Post	-0.045	0.002	-0.007	-0.046
	(0.422)	(0.968)	(0.873)	(0.388)
	[-0.391, 0.0683]	[-0.272, 0.119]	[-0.283, 0.112]	[-0.393, 0.0604]
N	129434	98117	94472	123876
Panel B: Outcome - Indicator for Families with A Child Under 15				
Policy Pressure × Post	0.042	0.025	0.024	0.036
	(0.586)	(0.742)	(0.725)	(0.603)
	[-0.303, 0.150]	[-0.297, 0.181]	[-0.298, 0.165]	[-0.314, 0.146]
N	129434	98117	94472	123876
Personal Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Destination City FE	Yes	Yes	Yes	Yes
Original Province FE	Yes	Yes	Yes	Yes

Note: Sample consists of families whose head is at prime age (20 to 55 years old), and has rural hukou. Unmarried families are dropped. We report wild bootstrap cluster p-values in parentheses and wild bootstrap cluster 95 percent confidence intervals in square brackets, generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at city level (11 clusters). *** significant at the 1 percent level; ** significant at the 5 percent level and * significant at the 10 percent level.

Table 10: The Effect of Migration Control Policy on Spousal Separation

Dependent Var: Spouse Separation				
Sample:	All Spouses	Spouses w/. A Child b/w 6-15	Spouses Less Educated	Spouses w/. A Child b/w 6-15, Less Educated
	(1)	(2)	(3)	(4)
Policy Pressure × Post	0.014	0.070	0.047	0.100
	(0.853)	(0.377)	(0.478)	(0.308)
	[-0.398, 0.305]	[-0.392, 0.406]	[-0.396, 0.275]	[-0.394, 0.404]
N	129434	62321	98117	44517
Personal Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Destination City FE	Yes	Yes	Yes	Yes
Original Province FE	Yes	Yes	Yes	Yes

Note: Spousal separation is an indicator taking value of one if spouses are not living together in the city, otherwise zero. We report wild bootstrap cluster p-values in parentheses and wild bootstrap cluster 95 percent confidence intervals in square brackets, generated using boottest command (10,000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at city level (11 clusters). *** significant at the 1 percent level; ** significant at the 5 percent level and * significant at the 10 percent level.

Conflict of Interest Statement

Both authors declare that they have no conflicts of interest.